



Information Leaflet: Enhance user focus in real world using XR

The study title is self-explanatory: Enhance user focus in real world using XR.

Invitation Paragraph

You are being invited to take part in a research study about how visual modifications in an Augmented Reality (AR) headset can help people stay focused in real-world situations.

Participation is voluntary.

Before you decide if you want to take part, it is important for you to understand why the research is being done and what taking part involves.

Please take time to read the following information carefully and discuss it with others if you wish. Please ask us if there is anything that is not clear or if you would like more information.

What is the purpose of the study?

We plan to study whether using shader-driven visual modifications (specifically desaturation, blur, and boundary highlighting) applied via AR passthrough on a Meta Quest headset can effectively guide and sustain user attention in real-world task contexts. The study aims to develop a Diminished Reality system that can improve task-relevant attention focus and reduce distractor salience.

The study is scheduled to run from the start of data collection on 25/05/2026 until the project end date of 14/08/2026.

For the purpose of this study, we will collect the following personal data:

- *Contact Information (Names and email addresses) for recruitment, scheduling, and obtaining informed consent.*
- *Performance Metrics (Task completion times and error rates).*



- *Attention & Tracking Data (Head direction/tracking data, time-to-first-fixation on distractors, and fixation distribution).*
- *Survey/Questionnaire Responses (Cognitive load scores via NASA-TLX, situational awareness responses, and user preference ratings via Likert scales).*

Why have I been chosen to participate?

We are interested in understanding the experiences of adults in using head-mounted Augmented Reality systems for attention guidance across various tasks (driving, classroom, and workstation scenarios).

You are being invited to participate because you are an adult (18 or over), have normal or corrected-to-normal vision, and are comfortable wearing a head-mounted display for approximately 45–60 minutes. You must not have any known history of epilepsy, photosensitive conditions, or serious visual impairment.

We are hoping to have 20 participants in the study.

Do I have to take part?

It is up to you to decide whether or not to take part. If you do decide to take part you will be given this information leaflet to keep and asked to sign a consent form.

If you decide to take part you are still free to change your mind at any time and without giving a reason. There will be no penalty or loss of benefits should you decide to withdraw.

What will happen to me if I take part?

This study involves attending a single session, lasting approximately 45–60 minutes, which will take place in a quiet, controlled laboratory environment on the Trinity College Dublin campus.

The session will include a 5-minute briefing and consent process, followed by three task scenarios (each lasting approximately 10 minutes) with and without the AR system.



You will complete post-condition questionnaires (NASA-TLX and Likert scales) to assess cognitive load and user preferences.

Data Handling:

- *All research data will be pseudonymised upon collection using a participant ID code, with the linking key stored separately and securely.*
- *Participants may request withdrawal of their data for up to two weeks after participation. During this period, the linking key will be retained securely to allow identification of participant data if withdrawal is requested.*
- *After the two-week withdrawal period, the linking key will be permanently deleted and the research dataset fully anonymised. After anonymisation, it will no longer be possible to identify or remove individual participant data.*
- *Contact information and signed consent forms will be stored securely on a password-protected university-managed drive accessible only to the researcher and supervisor, and retained in line with Trinity College Dublin postgraduate research retention guidelines.*
- *Fully anonymised research data may be retained for academic publication and research purposes.*

What are the benefits of taking part?

You will gain knowledge about your own attention dynamics and how peripheral distractions may affect your focus in different environments. The results of this study will help us to understand how customised Augmented Reality systems can be developed to act as attention coaches in real-world scenarios.

What are the possible disadvantages and risks of taking part?

We have taken all measures to minimise any risk to you. The primary risk is the potential for mild discomfort or cybersickness associated with wearing a Head-Mounted Display (HMD) for the 45–60 minute session.

In the event of you experiencing significant discomfort, motion sickness, or visual strain, the researcher will immediately stop the session at your request or if they observe signs of



distress. The researcher will advise you to rest. All information collected will be treated confidentially.

Will my participation in this study be kept confidential?

Your privacy is important to us. All digital data, including performance metrics and survey responses, will be pseudonymised immediately upon collection using a unique participant ID code. The data will be stored securely on Trinity College Dublin's official IT-provisioned cloud infrastructure (Microsoft OneDrive/SharePoint).

The code-key linking participant identities to participant ID codes will be retained securely for up to two weeks after participation to allow data withdrawal requests to be processed. After this period, the code-key will be permanently deleted and the dataset fully anonymised.

Any information that leaves Trinity, for example in a publication, will be fully anonymised so that your identity remains confidential. Anonymised, aggregated data may be shared with the scientific community and industry.

We will never share personal audio or video recordings with any third parties, as no such files are created during this study.

*It is **not** foreseeable that participants could potentially reveal information that the research team has a legal obligation to disclose (e.g., child protection policy, malpractice, or criminal activity). However, please be aware that researchers are legally obliged to report any such inadvertent discovery as required under relevant policy, legislation, or code of ethics.*

What will happen to the results of this research?

The research is being conducted as part of a Taught Masters program and the results will be included in the final dissertation. Fully anonymised data may also be published in future scientific papers or presented at conferences. If this is the case, your identity will remain confidential and no one will know that you took part in the study.

What do I do if I have any further questions?



Trinity College Dublin

Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

If you have any questions or concerns regarding your participation in the study please ask the researcher that gave you this information leaflet for further information. They will be happy to answer any questions that you may have.

THANK YOU

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The Information Leaflet should be dated and given a version number

One copy should be given to the individual. One copy should be retained on file by the research team